RMO(INDIA) – 2012(2)

Max Mark – 102 Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- Answer all the questions.
- All Questions carry equal marks.
- Answer to each questions starts on a new page. Clearly indicate the question number.

1. Let ABCD be a unit square. Draw a quadrant of a circle with A as centre and B, Das end points of the arc. Similarly draw a quadrant of a circle with B as centre and A, C as end points of the arc. Inscribe a circle Γ touching the arc AC internally, the arc BD internally and also touching the side AB. Find the radius of the circle Γ.
2. Let a, b and c are positive integers such that a divides b^4, b divides c^4 and c divides a^4**.** Prove that abc divides $(a+b+c)^{21}$.
3. Let a and b are positive real numbers such that $a+b=1$. Prove that $a^a b^b + a^b b^a \le 1$.
4. Let $X = \{1,2,3,4,5,6,7,8,9,10,11,12\}$. Find the number of pairs $\{A, B\}$ such that $A \subseteq B, B \subseteq X, A \ne B$ and $A \cap B = \{2,3,5,7,8\}$
5. Let ABC be a triangle. Let D, E be a points on the segment BC such that BD = DE = EC. Let F be the mid point of AC. Let BF intersect AD in P and AE in Q respectively. Determine BP/PQ.
6. Show that for all real numbers x, y, z such that $x+y+z=0$ and $xy+yz+zx=-3$, the expression $x^3y+y^3z+z^3x$ is a constant.

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